scRNA-seq with Seurat

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Walk through the Seurat pipeline: QC, normalisation, feature selection, PCA, neighbour graph, clustering, UMAP.
- Describe why scRNA-seq counts need different handling from bulk counts.
- Produce a small analogous pipeline in base R with simulated cells and the `uwot` UMAP.

Prerequisites

UMAP; bulk RNA-seq differential expression.

Background

Single-cell RNA-seq measures gene expression in thousands of cells simultaneously, producing a sparse, noisy, zero-inflated matrix. Clustering and cell-type annotation are exploratory steps that require a chain of careful preprocessing: drop empty droplets, normalise library size, select variable genes, reduce to principal components, build a nearest-neighbour graph, cluster on the graph, embed with UMAP.

Seurat is the dominant R toolkit for this pipeline. Every step has alternatives — `SCTransform` vs `NormalizeData`, Louvain vs Leiden clustering, Harmony for batch correction — and each choice changes the resulting partition. The good reflex is to run the same pipeline twice with a perturbed parameter and check that the biology survives.

Seurat is not on CRAN and its installation is non-trivial in a rendering environment. We sketch the pipeline with `#| eval: false` and run an analogous, tiny base-R + `uwot` pipeline on a simulated cell × gene matrix so the reasoning is visible in every render.

Setup

```r
library(tidyverse)
library(uwot)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Walk through an scRNA-seq analysis conceptually with a small simulated cell × gene matrix, ending at a 2-D embedding with cluster labels.

2. Approach

300 cells, 500 genes, three cell types with partly overlapping expression programmes.

```r
cell_type <- rep(c("T", "B", "myeloid"), each = n_cell / 3)
mu_base <- exp(rnorm(n_gene, 1, 1))
X <- matrix(0, n_cell, n_gene)
```

```r
for (i in seq_len(n_cell)) {
  program <- rep(1, n_gene)
  if (cell_type[i] == "T")       program[1:20]  <- 5
  if (cell_type[i] == "B")       program[21:40] <- 5
  if (cell_type[i] == "myeloid") program[41:60] <- 5
  X[i, ] <- rnbinom(n_gene, mu = mu_base * program, size = 2)
}
tibble(lib = rowSums(X), ct = cell_type) |>
  ggplot(aes(ct, lib)) + geom_boxplot() +
  labs(x = NULL, y = "total counts per cell")
```

3. Execution

Log-normalise, select variable genes, PCA, UMAP, cluster.

```r
gene_var <- apply(logX, 2, var)
keep <- order(gene_var, decreasing = TRUE)[1:100]
pcs <- prcomp(logX[, keep])$x[, 1:10]
emb <- umap(pcs, n_neighbors = 20, min_dist = 0.1)

# A simple k-means cluster on the PCs as a stand-in.
km <- kmeans(pcs, centers = 3, nstart = 10)
tibble(x = emb[, 1], y = emb[, 2],
       cluster = factor(km$cluster), truth = cell_type) |>
  ggplot(aes(x, y, colour = cluster, shape = truth)) +
  geom_point(alpha = 0.8, size = 1) +
  labs(x = "UMAP1", y = "UMAP2")
```

A Seurat call pattern (not run).

```r
library(Seurat)
obj <- CreateSeuratObject(counts = t(X))
obj <- NormalizeData(obj)
obj <- FindVariableFeatures(obj, nfeatures = 100)
obj <- ScaleData(obj)
obj <- RunPCA(obj, npcs = 10)
obj <- FindNeighbors(obj, dims = 1:10)
obj <- FindClusters(obj, resolution = 0.5)
obj <- RunUMAP(obj, dims = 1:10)
DimPlot(obj, label = TRUE)
```

4. Check

Contingency of cluster vs true cell type.

5. Report

> On a simulated 300-cell × 500-gene dataset with three cell types, a small pipeline (log normalisation, top-100 variable genes, 10 PCs, UMAP, k-means) recovered the three types with high agreement. The same reasoning — preprocess, reduce, graph, cluster, embed — underlies the Seurat pipeline sketched in the accompanying code.

On a real dataset the critical choices are the normalisation (`SCTransform` vs `NormalizeData`), the number of PCs (elbow or `JackStraw`), and the clustering resolution. Perturb each and confirm that the biological conclusions do not.

Common pitfalls

- Clustering before QC; apoptotic cells and empty droplets cluster together.
- Forgetting that UMAP distances are not biological distances.
- Using marker genes from a tutorial on a different tissue for cell-type annotation.

Further reading

- Stuart T et al. (2019), *Comprehensive integration of single-cell data.*
- Luecken MD, Theis FJ (2019), *Current best practices in single-cell RNA-seq analysis: a tutorial.*

Session info

```
```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)